

EEG Replication Study Report: P600 Amplitude Extraction and Sentence Norming Correlation Analysis

1. Project Overview and Scientific Objective

This report documents the complete data processing, methodological troubleshooting, and statistical analysis workflow for a large-scale electroencephalography (EEG) replication study based on the DeLong et al. (2005) paradigm (utilizing the Nieuwland et al. multi-laboratory OSF dataset). The experiment investigated phonological and semantic prediction during rapid serial visual presentation (RSVP) sentence reading, examining neural responses to English indefinite articles ("a" or "an"), the preceding target verbs, and the subsequent expected or unexpected nouns.

The primary objective of the current analysis was to test whether the mean amplitude of the P600 event-related potential (ERP) component, elicited by the verb preceding the article, correlates positively with the sentence constraint and noun expectancy (Cloze Probability / Norming score) across 80 experimental sentences. The complete multi-site dataset spans 9 university laboratories across the United Kingdom (BIRM, BRIS, EDIN, GLAS, KENT, LOND, OXFO, STIR, YORK), comprising 352 participants and 157,320 extracted electrode-level trial observations.

2. EEG Preprocessing and Cleaning Pipeline (folder "matlab codes")

Raw EEG recordings from all 9 laboratories were preprocessed in MATLAB using the EEGLAB toolbox under a standardized, automated artifact-rejection protocol (saved as *_cleaned.set files in "processed data - cleansed files" folder):

- **Bandpass Filtering and Re-referencing:** Continuous EEG signals were bandpass-filtered at **0.1–40 Hz** (pop_eegfiltnew) to attenuate slow DC drifts and high-frequency muscle or line-noise artifacts. Data were re-referenced offline according to laboratory-specific setups (e.g., mastoids M1/M2, A1/A2, EXG1/EXG2, or TP9/TP10). For laboratories using high-density arrays or high sampling rates (e.g., OXFO, GLAS, LOND, YORK), data were downsampled to 250 Hz, and extraneous channels were removed to standardize the spatial layout.
- **Independent Component Analysis (ICA) and ICLabel:** ICA decomposition (runica) was performed on each subject dataset to isolate ocular artifacts (eye blinks and horizontal saccades) and localized noise. Ocular components were automatically identified and subtracted using the **ICLabel plugin, with a rejection threshold set at ≥60% probability** for the 'Eye' category (pop_icflag).

- **RSVP Presentation Timing, Trigger Selection, and Epoching:** Epochs were time-locked ($t = 0$ ms) to the onset of the indefinite article ("a"/"an"), marked by trigger codes 201 (s201/S201/201, Violation_201) and 202 (s202/S202/202, Control_202). Baseline correction was applied over the [-1000, -500] ms interval, corresponding to the word preceding the verb.
- **Extreme Value Thresholding:** An automated voltage threshold rejection (pop_eegthresh) was applied specifically to the 12 central-posterior Region of Interest (ROI) electrodes (Pz, Cz, CP1, CP2, P3, P4, C3, C4, P7, P8, O1, O2). Epochs containing voltage fluctuations exceeding $\pm 100 \mu\text{V}$ within the -100 to 490 ms window were rejected to remove remaining physical artifacts prior to final analysis.

3. Analysis Table Construction, Trial-to-Item Reconciliation, and Multi-Lab Integration

To evaluate the correlation between P600 amplitude and norming scores from the stimulus master table (REPLICATION_ITEMS.xlsx, containing 80 items across the Delong_Replication sheet), each preprocessed EEG epoch across all 9 laboratories had to be accurately mapped to its chronological Trial number and subsequently cross-referenced with behavioral log files to recover its true randomized stimulus Item Number (1–80). This reconciliation and data integration pipeline was executed in three sequential stages:

- Stage 1: Global EEG Extraction and True Chronological Trial Identification (P600_extractor_correct_trial_number.m in matlab codes folder): Initial extraction scripts recorded the sequential EEGLAB epoch loop index ($ep = 1:\text{EEG.trials}$) rather than the true trial identifier. Because artifact rejection removes noisy epochs and sentence presentation order was randomized across participants, relying on the surviving epoch index misaligned trials across datasets. Furthermore, attempting to search backward across continuous EEG.urevent structures for raw 1–80 markers caused trigger bleeding across trial boundaries. Inspection of the EEG event structure (EEG.epoch(ep).eventtype) confirmed that triggers 101 to 180 (encoded as numeric values or strings s101–s180 / S101–S180) represent the chronological trial index (1–80) rather than the stimulus identity. Using the MATLAB extraction script (P600_extractor_correct_trial_number.m), all cleaned datasets (*_cleaned.set) across all 9 laboratories (BIRM, BRIS, EDIN, GLAS, KENT, LOND, OXFO, STIR, YORK) were processed to extract mean P600 amplitudes over the [0, 500] ms window across the 12 ROI channels. For each epoch, the script identified the experimental condition (Violation_201 vs. Control_202) and stripped the 's'/'S' prefix from triggers 101–180, subtracting 100 (num_val - 100) to recover the exact chronological Trial number (1–80). All extracted observations were compiled into a unified multi-laboratory master file (Master_P600_Data_correct_trial_number.csv).
- Stage 2: Laboratory-Specific Behavioral Log Cross-Referencing (item_number_extracting.py in main folder): Because sentence order was

randomized per participant, the chronological Trial number (1–80) in Master_P600_Data_correct_trial_number.csv had to be matched to the actual stimulus Item Number (1–80) using each laboratory's behavioral log files. A dedicated Python pipeline (item_number_extracting.py) processed the two distinct behavioral logging software formats used across sites:

- Presentation (.log) Laboratories (BRIS, GLAS, LOND, OXFO, YORK): The script parsed individual tab-delimited .log files (dynamically locating the Subject header row where necessary), filtered for stimulus presentation events (Event Type == 'Picture'), and extracted each participant's Subject_ID. The true stimulus Item Number (Itemnumber(str)) and chronological trial number (Trialnumber(str), stored as Trial_Computed) were extracted and deduplicated at the sentence level.
- E-Prime (.txt / .emrg2) Laboratories (BIRM, STIR, EDIN, KENT): For BIRM (birm_log_files_merged.txt) and STIR (stir_logs_merged.txt), the script read the merged UTF-16 tabular E-Prime exports, whereas for EDIN and KENT it parsed individual raw E-Prime .txt log files by iterating through trial log frames (`*** LogFrame Start ***` to `*** LogFrame End ***`). Because E-Prime logs recorded within-block item indices (1–20) across four experimental block columns (Blocks1–Blocks4), the true global Item Number (1–80) was reconstructed using the formula $((i - 1) * 20) + \text{val}$ for block i . Practice trials were excluded by filtering for rows with valid Item Number and word presentation timestamps (RSVP.OnsetTime / RSVP1–3.OnsetTime), and experimental trials were sorted chronologically by OnsetTime within each Subject_ID to compute the sequential chronological trial index (Trial_Computed = 1 to 80).
- Per-Lab Table Generation: For each of the 9 laboratories, the EEG subset from Master_P600_Data_correct_trial_number.csv was inner-joined with the behavioral log mapping on ['Subject_ID', 'Trial'] == ['Subject_ID', 'Trial_Computed'], merged with the stimulus norming table (REPLICATION_ITEMS.xlsx, column H) on Item Number, and exported as an individual laboratory analysis table (Final_Analysis_Table_BIRM.csv through Final_Analysis_Table_YORK.csv).
- Stage 3: Multi-Laboratory Aggregation and Correlation Analysis (correlation_analysis.py in main folder): In the final stage, all 9 laboratory-specific tables (Final_Analysis_Table_<LAB>.csv) were loaded and concatenated into a single unified dataset (all_data) using correlation_analysis.py. After removing missing values (NaN), item-level mean P600 amplitudes (Mean_Amplitude) were aggregated across participants and evaluated against stimulus Norming scores across four analytical tiers: (1) individual scatter plots for each of the 9 laboratories separately; (2) a combined linear regression averaging across all 9 laboratories and all 12 ROI electrodes; (3) an electrode-by-electrode regression grid (FacetGrid) across all 12 channels; and (4) a targeted regression restricted strictly to the parietal electrode cluster (P3, P4, P7, P8, Pz). The final cross-laboratory item-level dataset was exported to Combined_Averages_Summary.csv.

4. Multi-Laboratory Correlation Analysis

Using the consolidated data from all 9 laboratories, a comprehensive correlation analysis was executed via a dedicated Python pipeline (`correlation_analysis.py`) to evaluate the relationship between the norming scores and the elicited P600 mean amplitudes. To thoroughly investigate potential trends, the data was aggregated and visualized across four analytical tiers:

- **Individual Laboratory Level:** Separate scatter plots were generated for each of the 9 university datasets to identify any localized or site-specific variance.
- **Global Average (All Electrodes):** A combined linear regression evaluated the mean P600 amplitude against the norming score for each experimental item, averaging across all participants, laboratories, and the 12 region-of-interest (ROI) electrodes.
- **Electrode-Specific Grid:** A comprehensive facet grid of regression models was generated to assess the correlation independently within each of the 12 recording channels.
- **Targeted Parietal ROI:** A focused regression analysis was conducted exclusively on the parietal electrode cluster (P3, P4, P7, P8, Pz), isolating the spatial region where the P600 component is typically most pronounced.

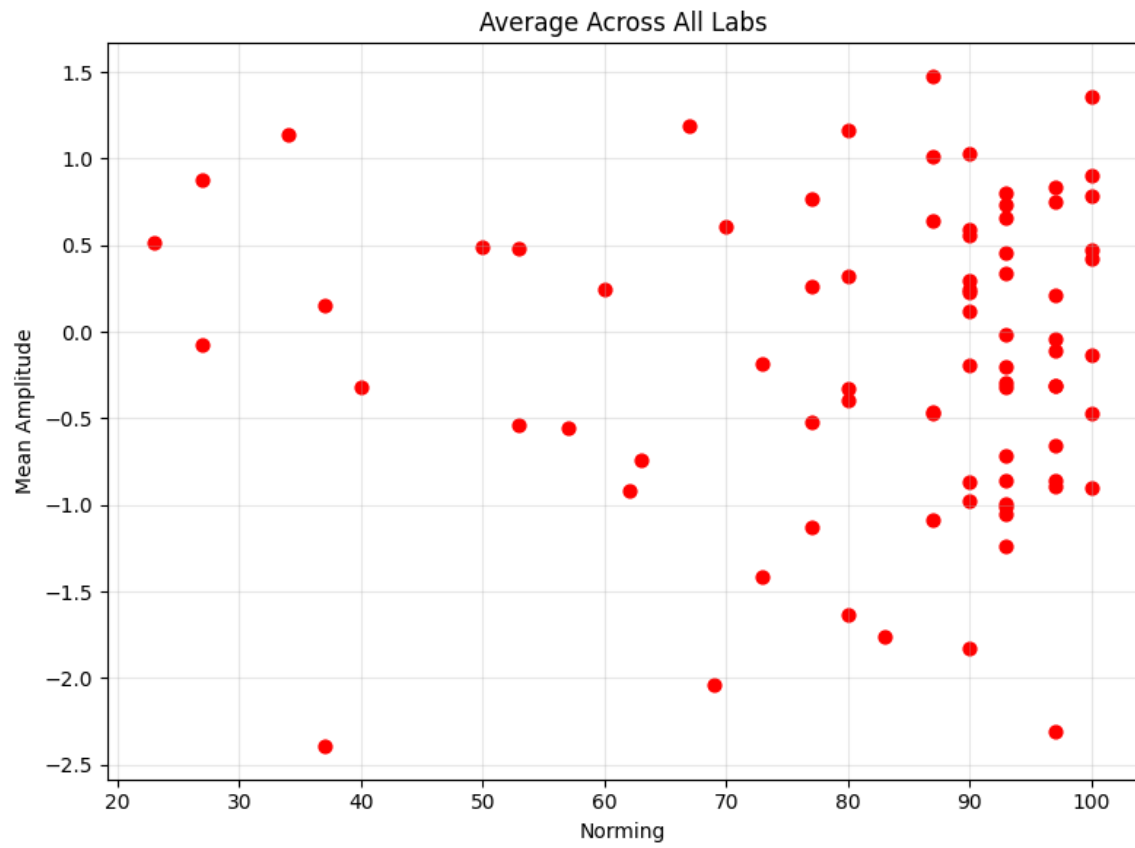
Following these computations, the cross-laboratory item-level averages were exported as a final summary table (`Combined_Averages_Summary.csv`).

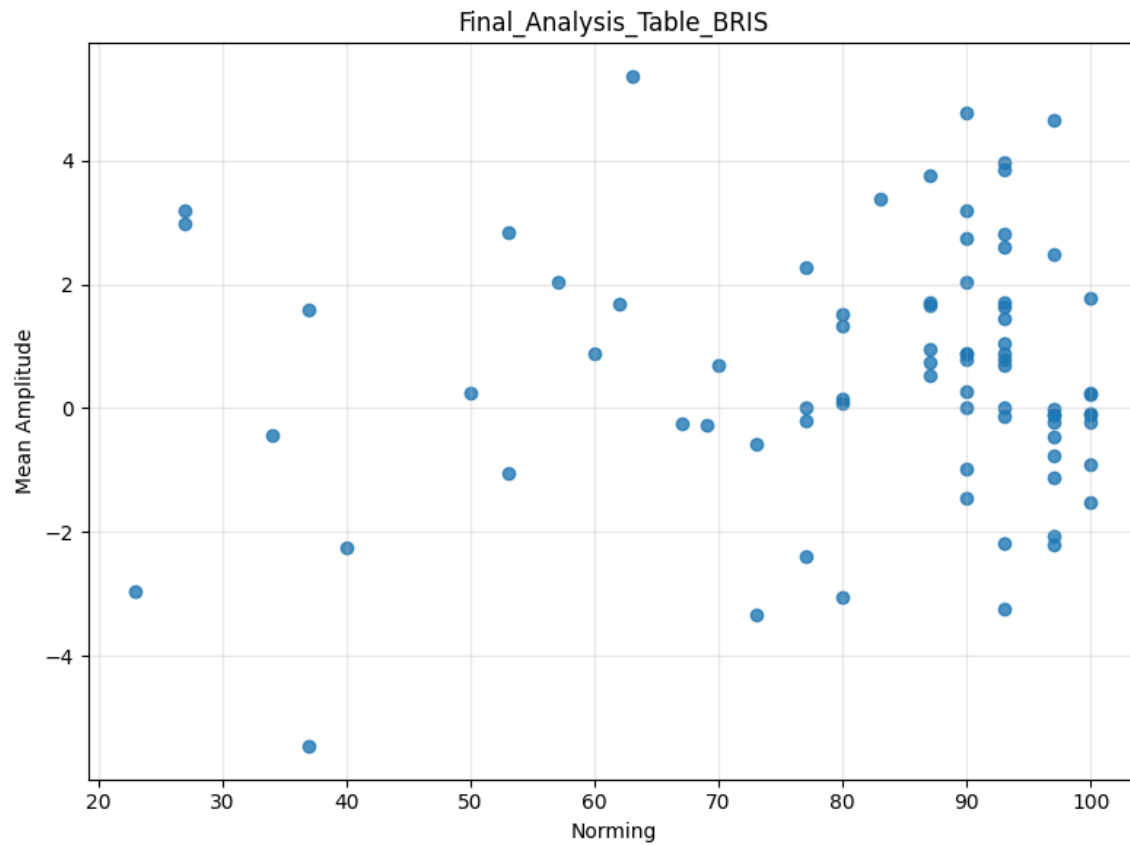
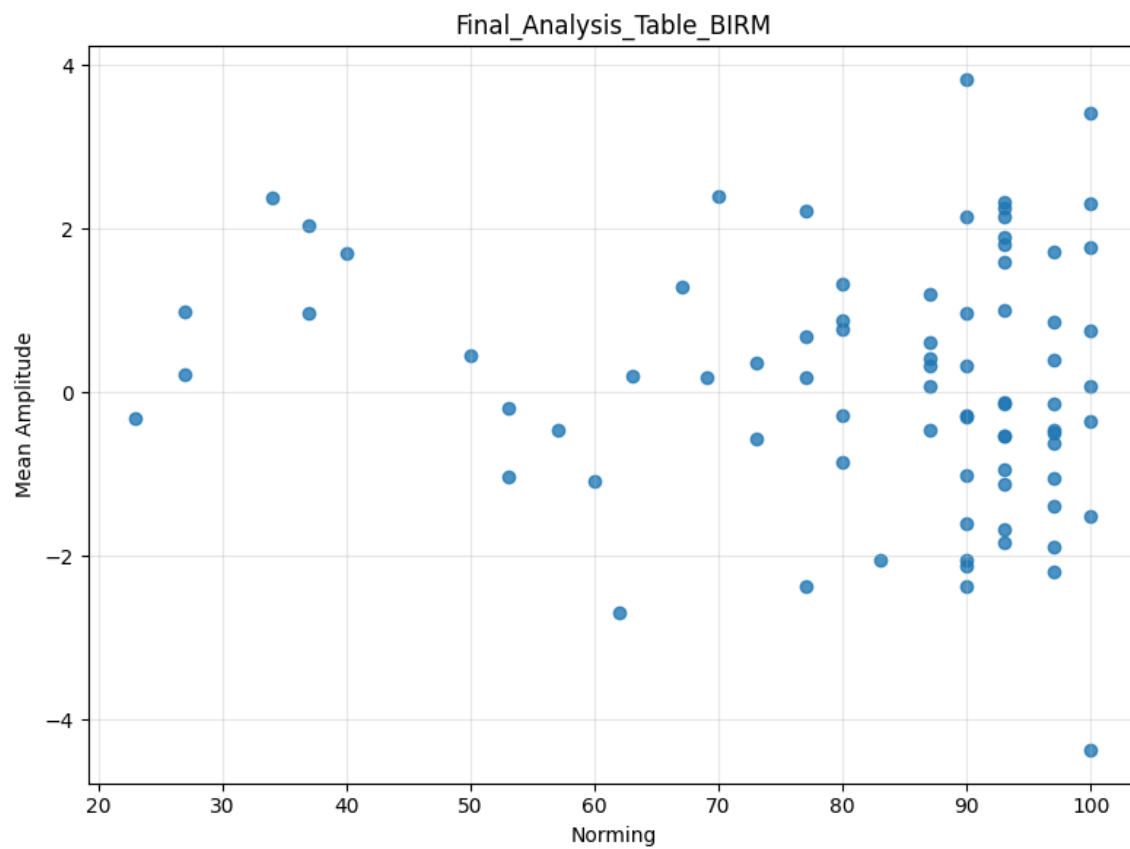
5. Results and Methodological Explanations for the Null Linear Correlation

Across all executed analyses and data subsets, the results indicated a consistent lack of clear linear correlation between the norming scores and the P600 mean amplitudes. No significant correlation emerged at the individual laboratory level, the global average level, the individual electrode level, or within the targeted parietal cluster. This comprehensive null result is primarily driven by the structural properties of the experimental stimuli:

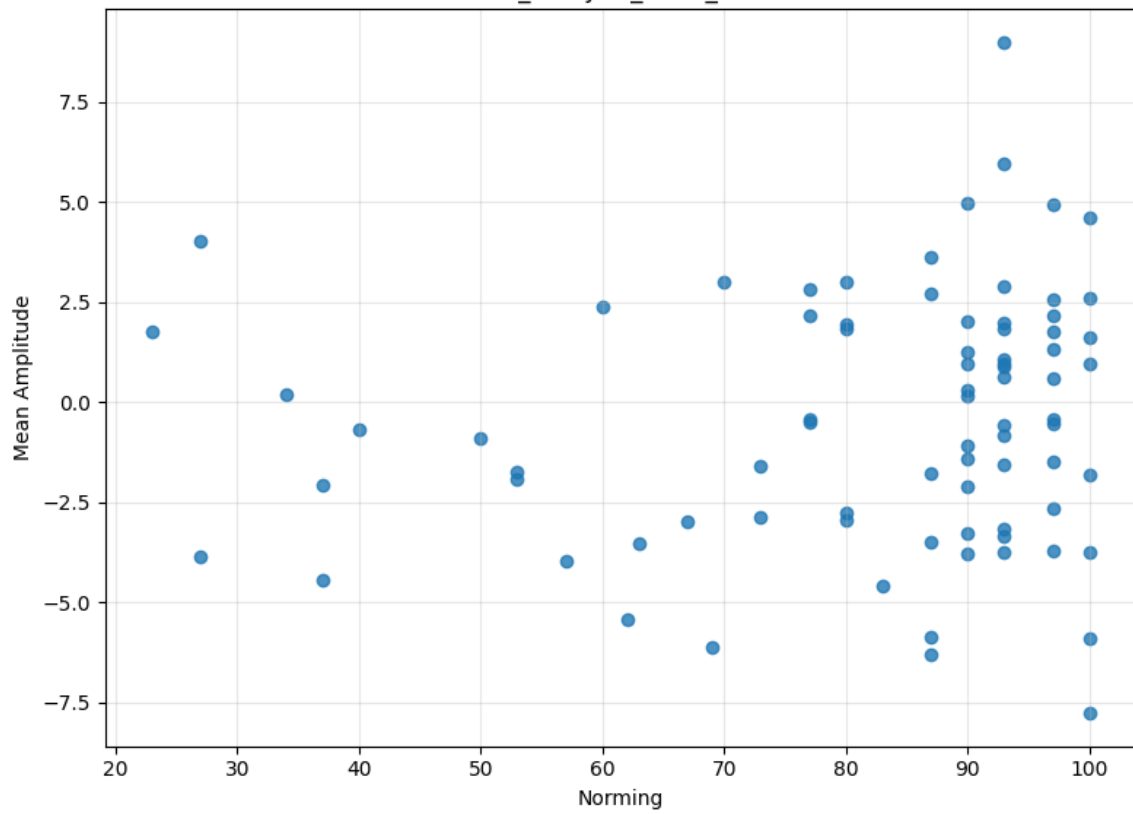
- **Bimodal Distribution of Expectancy:** By experimental design, the noun norming scores exhibit a strict bimodal distribution. Violation condition items cluster near zero (ranging from 0.000% to 34.000%), whereas control condition items cluster near the ceiling (ranging from 62.000% to 100.000%).
- **Violation Effect vs. Linear Trend:** There are no intermediate norming values between 34% and 62%. While violation trials successfully elicited higher mean P600 amplitudes compared to control trials, plotting this categorical violation effect against raw expectancy percentages inherently produces a slightly negative slope rather than a continuous linear correlation.

6. Final graphs from correlation_analysis.py

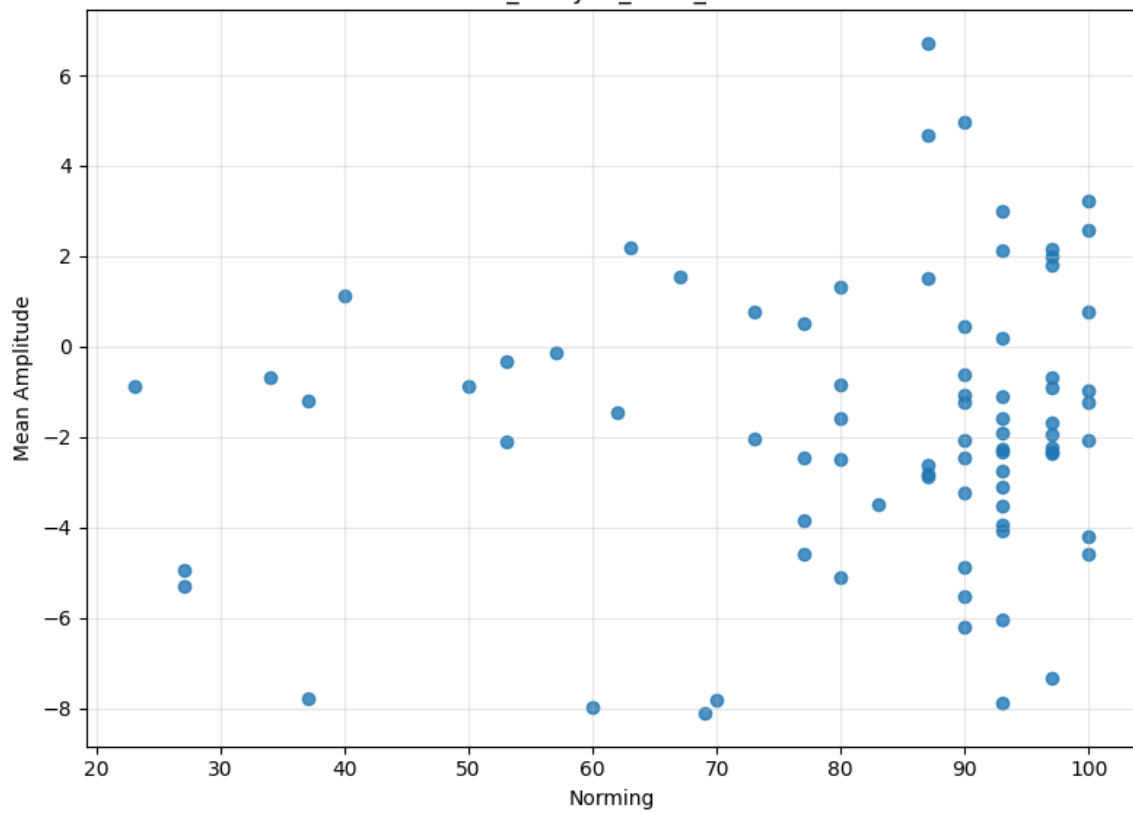


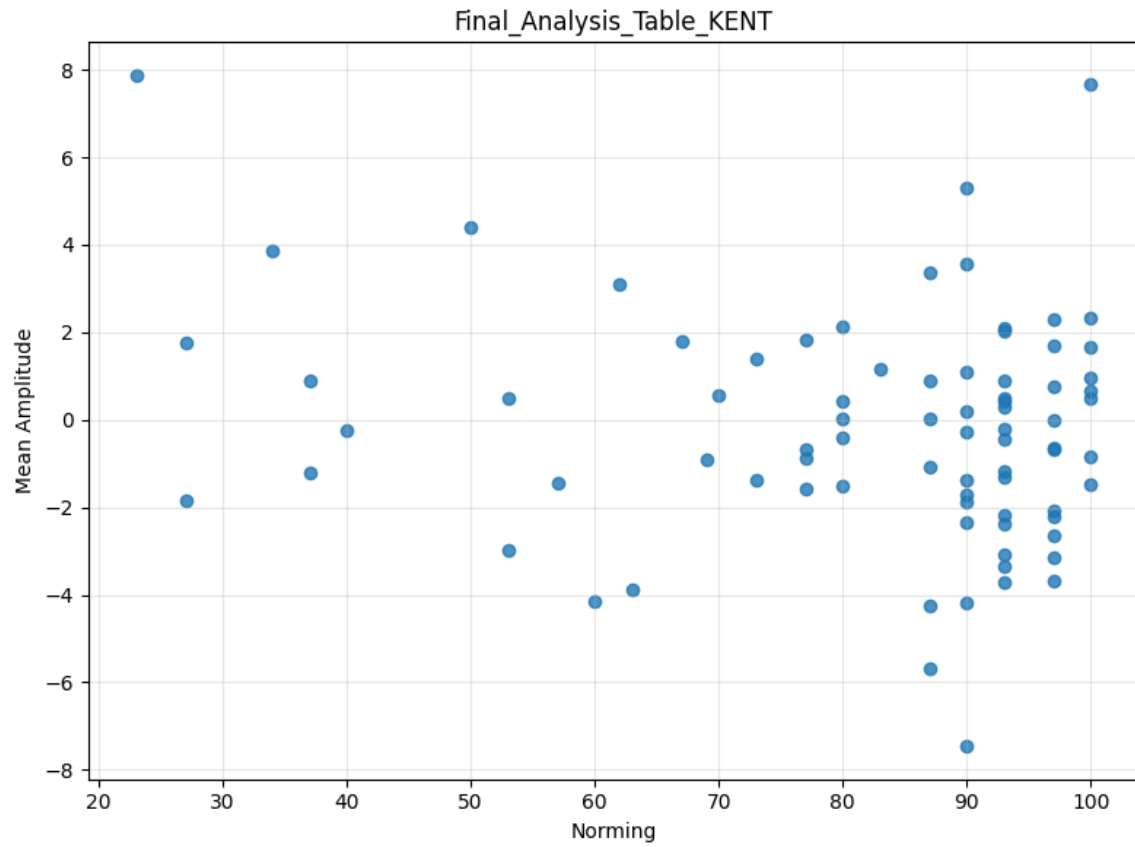


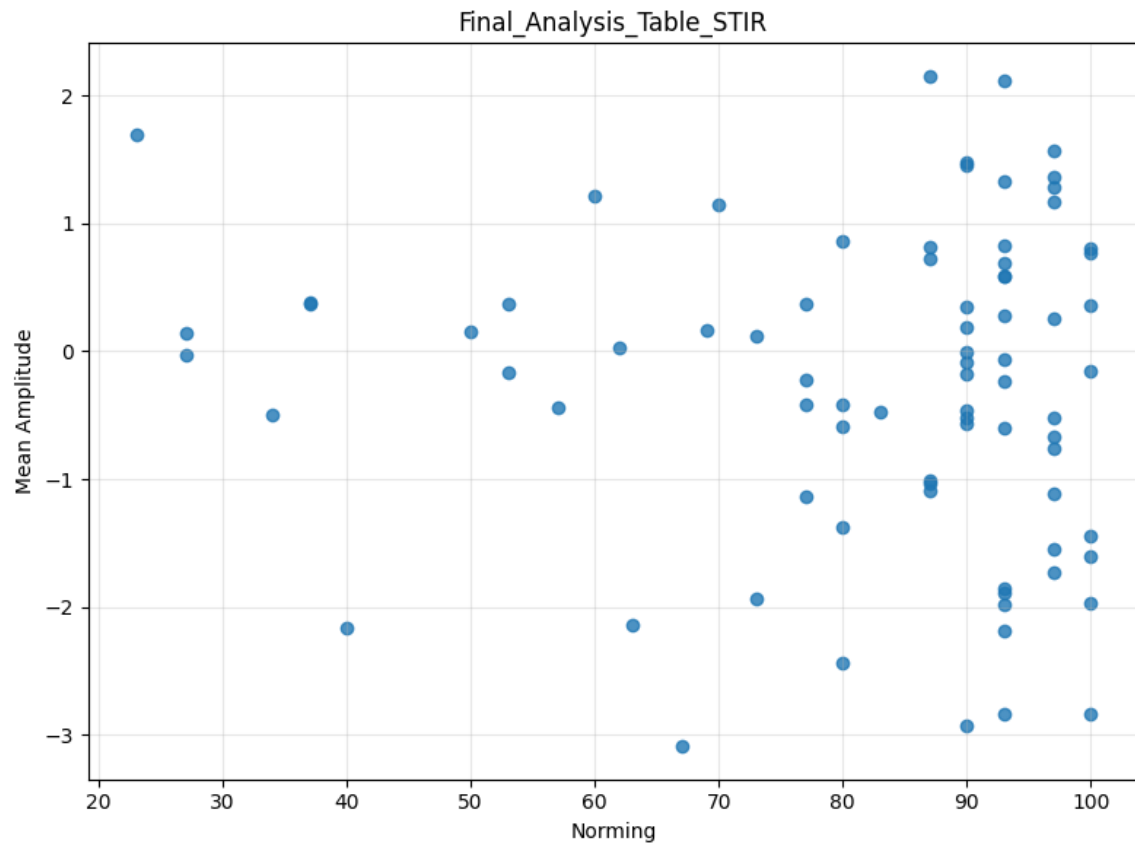
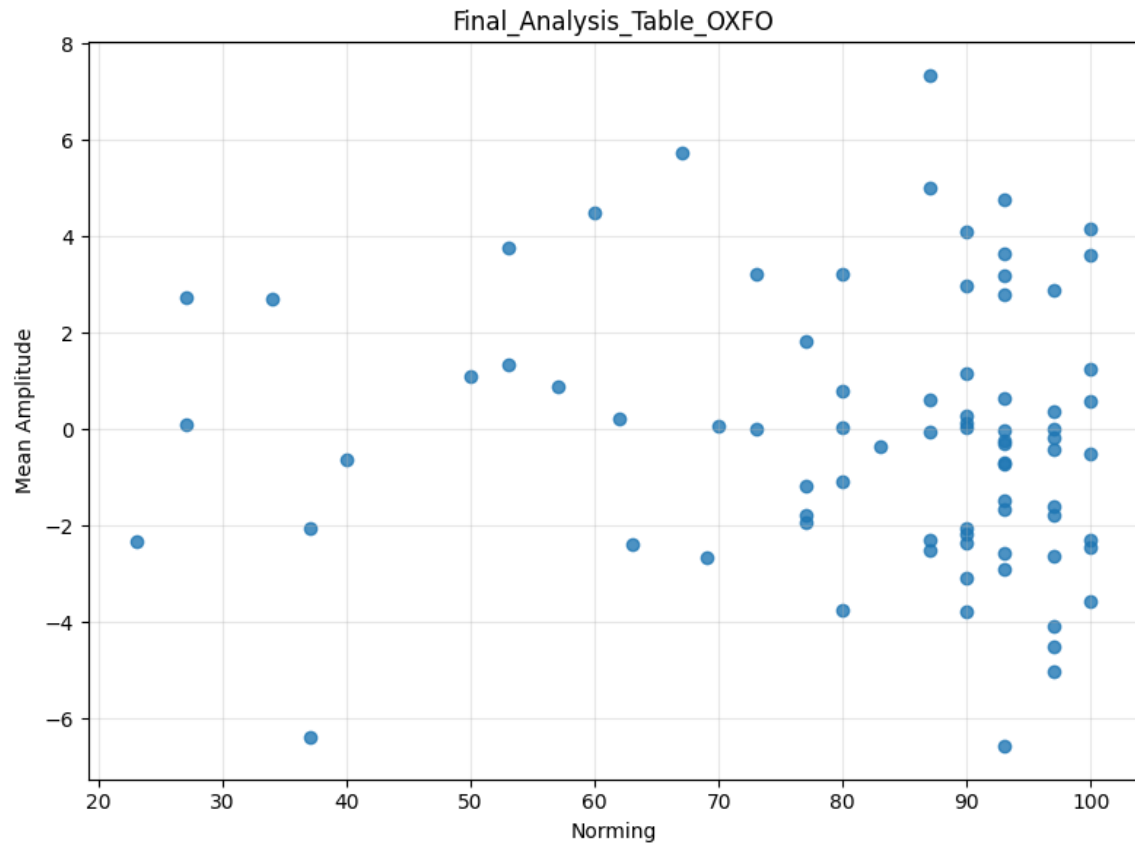
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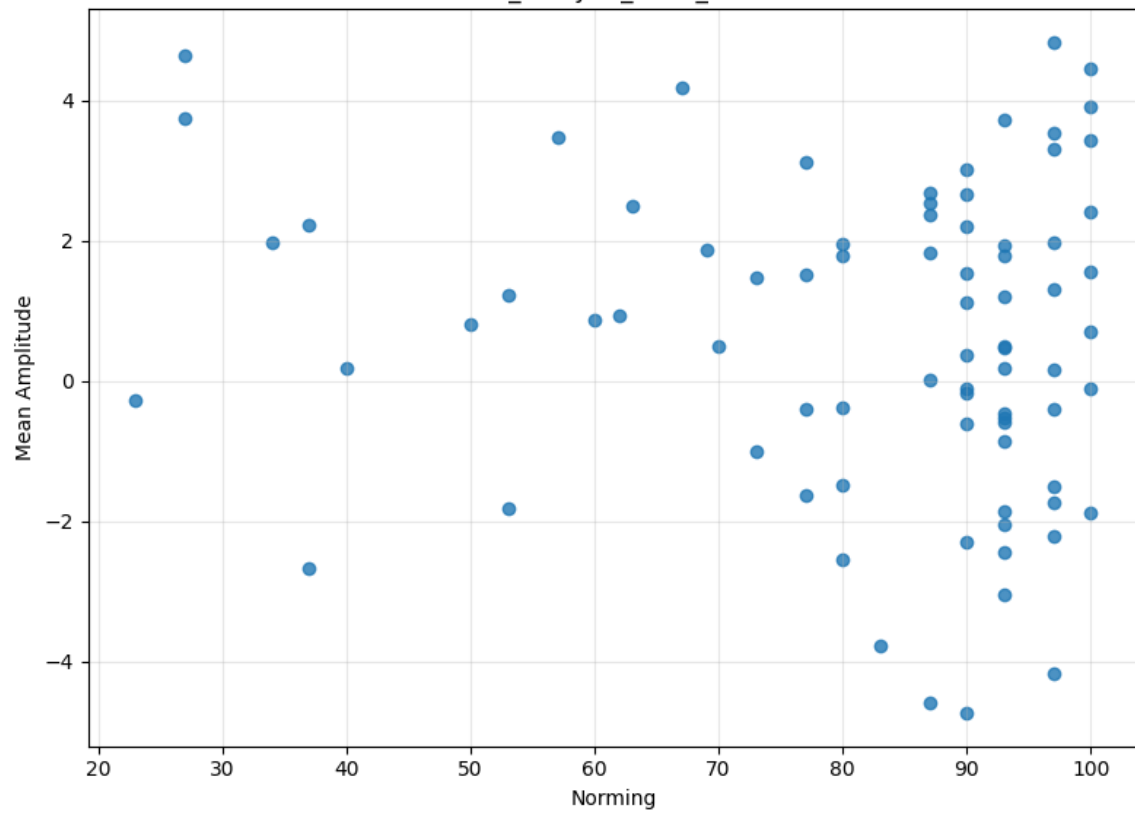
Final_Analysis_Table_GLAS

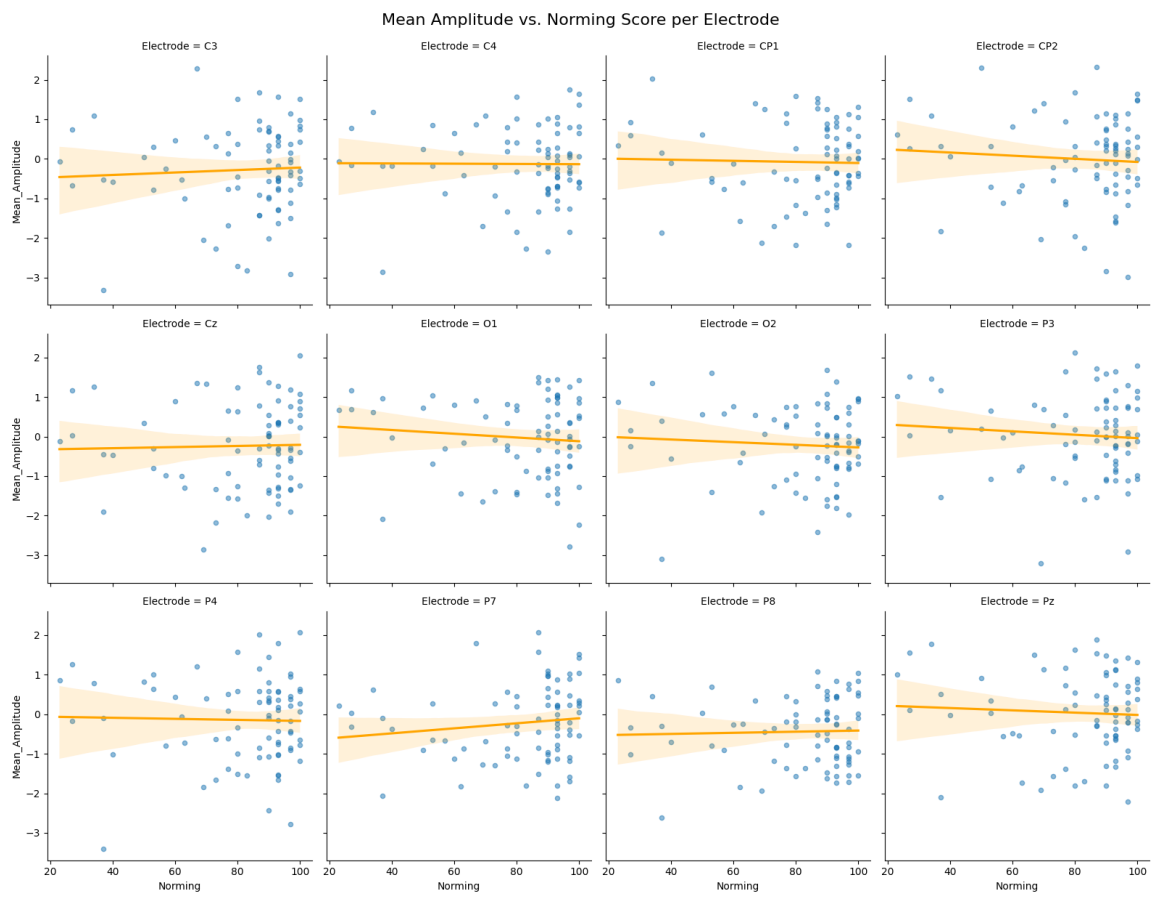
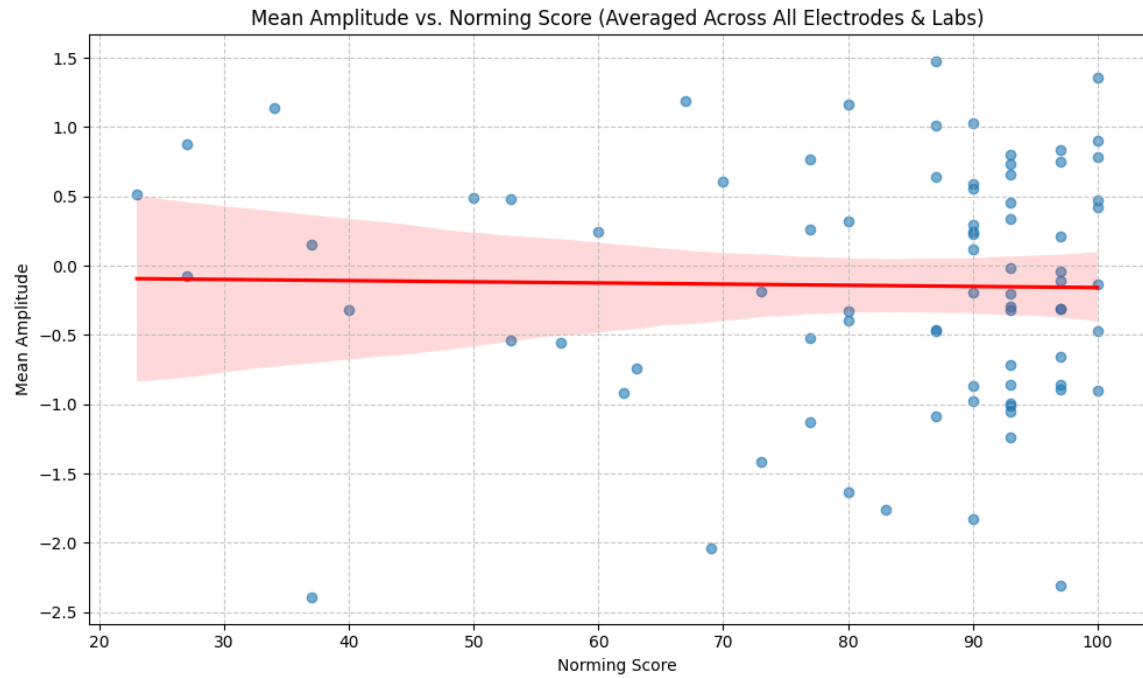






Final_Analysis_Table_YORK





Mean Amplitude vs. Norming Score (Parietal Electrodes Only: P3, P4, P7, P8, Pz)

